Understanding Lab 5 and Lab 6

Hill Cipher, One Time Pad, and Affine Cipher

AI 331 — Information Security and Cryptography, SVNIT Surat

5th Sem B.Tech. (AI)

Before the ciphers: letters as numbers

Every cipher in these two labs treats a letter as a number:

Letter	A	B	C	D	E	F	G	H	I	J	K	L	M
Number	0	1	2	3	4	5	6	7	8	9	10	11	12

Letter	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Number	13	14	15	16	17	18	19	20	21	22	23	24	25

All arithmetic is **mod 26**. That means: do the calculation, then keep only the remainder after dividing by 26. The remainder is always between 0 and 25, so it is always a letter again.

Example: $48 \bmod 26 = 22$, because $26 \times 1 = 26$ and $48 - 26 = 22$. Number 22 is the letter **W**.

If a subtraction goes negative, add 26 until it is positive. Example: $4 - 23 = -19$, and $-19 + 26 = 7$, which is **H**.

Spaces are removed. Only letters are encrypted.

Lab 5 — Hill Cipher (2×2)

What it is

Hill Cipher does **not** encrypt one letter at a time. It encrypts **two letters together**, using a 2×2 key matrix.

Given in the assignment:

- Key matrix $K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$
- Plaintext: **MEET AT BASE**

Step 1. Write the key matrix

$$K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

Top row is 3, 3. Bottom row is 2, 5. This matrix is the whole key.

Step 2. Clean the plaintext and convert to numbers

Remove spaces and use capital letters:

MEET AT BASE becomes **MEETATBASE**

Letter	M	E	E	T	A	T	B	A	S	E
Number	12	4	4	19	0	19	1	0	18	4

If the length were odd, we would add an extra **X** so every block has two letters. Here the length is 10, so no padding is needed.

Step 3. Split into pairs

Block	ME	ET	AT	BA	SE
Numbers	[12, 4]	[4, 19]	[0, 19]	[1, 0]	[18, 4]

Each pair is a column vector $\begin{bmatrix} p_1 \\ p_2 \end{bmatrix}$.

Step 4. Encrypt with matrix multiplication

$$\begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} p_1 \\ p_2 \end{bmatrix} \pmod{26}$$

Which means:

$$c_1 = (3p_1 + 3p_2) \bmod 26$$

$$c_2 = (2p_1 + 5p_2) \bmod 26$$

Then convert c_1 and c_2 back to letters.

Block ME = [12, 4]

$$c_1 = 3(12) + 3(4) = 36 + 12 = 48, \quad 48 \bmod 26 = 22 \rightarrow \mathbf{W}$$

$$c_2 = 2(12) + 5(4) = 24 + 20 = 44, \quad 44 \bmod 26 = 18 \rightarrow \mathbf{S}$$

ME → WS

Block ET = [4, 19]

$$c_1 = 3(4) + 3(19) = 12 + 57 = 69, \quad 69 \bmod 26 = 17 \rightarrow \mathbf{R}$$

$$c_2 = 2(4) + 5(19) = 8 + 95 = 103, \quad 103 \bmod 26 = 25 \rightarrow \mathbf{Z}$$

ET → RZ

Block AT = [0, 19]

$$c_1 = 3(0) + 3(19) = 57, \quad 57 \bmod 26 = 5 \rightarrow \mathbf{F}$$

$$c_2 = 2(0) + 5(19) = 95, \quad 95 \bmod 26 = 17 \rightarrow \mathbf{R}$$

AT → FR

Block BA = [1, 0]

$$c_1 = 3(1) + 3(0) = 3 \rightarrow \mathbf{D}$$

$$c_2 = 2(1) + 5(0) = 2 \rightarrow \mathbf{C}$$

BA → DC

Block SE = [18, 4]

$$c_1 = 3(18) + 3(4) = 54 + 12 = 66, \quad 66 \bmod 26 = 14 \rightarrow \mathbf{O}$$

$$c_2 = 2(18) + 5(4) = 36 + 20 = 56, \quad 56 \bmod 26 = 4 \rightarrow \mathbf{E}$$

SE → OE

Step 5. Ciphertext

Join the blocks:

WSRZFRDCOE

Plain block	ME	ET	AT	BA	SE
Cipher block	WS	RZ	FR	DC	OE

Step 6. Inverse of the key matrix (needed to decrypt)

Decryption uses K^{-1}, the inverse of K modulo 26. For a 2×2 matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$:

1. Determinant: $\det = ad - bc$
2. Reduce det modulo 26
3. Find the modular inverse of det (a number x such that $\det \cdot x \equiv 1 \pmod{26}$)
4. Multiply the adjugate by that inverse

For our key, $a = 3$, $b = 3$, $c = 2$, $d = 5$:

$$\det = (3)(5) - (3)(2) = 15 - 6 = 9$$

Now find x with $9x \equiv 1 \pmod{26}$. Try $x = 3$: $9 \times 3 = 27$, and $27 \bmod 26 = 1$. So the inverse of 9 is **3**.

The adjugate swaps a and d, and flips the signs of b and c:

$$\begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \begin{bmatrix} 5 & -3 \\ -2 & 3 \end{bmatrix}$$

Multiply every entry by 3, then reduce mod 26:

$$\begin{bmatrix} 5 \times 3 & -3 \times 3 \\ -2 \times 3 & 3 \times 3 \end{bmatrix} = \begin{bmatrix} 15 & -9 \\ -6 & 9 \end{bmatrix}$$

Negative values: $-9 + 26 = 17$, and $-6 + 26 = 20$.

$$K^{-1} = \begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix} \pmod{26}$$

This inverse exists only because $\gcd(9, 26) = 1$. If the determinant shared a factor with 26, the matrix could not be used as a Hill key.

Step 7. Decrypt

$$\begin{bmatrix} p_1 \\ p_2 \end{bmatrix} = \begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} \pmod{26}$$

Check the first cipher block **WS** = [22, 18]:

$$p_1 = 15(22) + 17(18) = 330 + 306 = 636, \quad 636 \bmod 26 = 12 \rightarrow \mathbf{M}$$

$$p_2 = 20(22) + 9(18) = 440 + 162 = 602, \quad 602 \bmod 26 = 4 \rightarrow \mathbf{E}$$

WS $\rightarrow$ **ME**. The other blocks work the same way and recover **MEETATBASE**.

How to remember it

- Encrypt: multiply the pair by K, then mod 26.
- Decrypt: multiply the pair by K^{-1}, then mod 26.
- Same matrix for the whole message. Changing one plaintext letter changes both ciphertext letters in its pair.

Lab 6, Question 1 — One Time Pad

What it is

A one-time pad encrypts **each letter with its own key letter**. The key must be:

- as long as the message
- random
- used only once

If those three things are true, the cipher cannot be broken, because every ciphertext is equally possible.

Formula

$$C = (P + K) \bmod 26$$

$$P = (C - K) \bmod 26$$

P is the plaintext letter's number, K is the key letter's number, C is the ciphertext letter's number.

Worked example

Plaintext: **HELLO**
Key: **XMCKL**

	1	2	3	4	5
Plain	H = 7	E = 4	L = 11	L = 11	O = 14
Key	X = 23	M = 12	C = 2	K = 10	L = 11
Sum	30	16	13	21	25
mod 26	4	16	13	21	25
Cipher	**E**	**Q**	**N**	**V**	**Z**

Ciphertext: **EQNVZ**

Check the first letter by hand: $7 + 23 = 30$, and $30 - 26 = 4$, which is **E**.

Decrypt the same example

Subtract the key, then fix negatives by adding 26.

Cipher	E = 4	Q = 16	N = 13	V = 21	Z = 25
Key	X = 23	M = 12	C = 2	K = 10	L = 11
$C - K$	$4 - 23 = -19$	$16 - 12 = 4$	$13 - 2 = 11$	$21 - 10 = 11$	$25 - 11 = 14$
mod 26	$-19 + 26 = 7$	4	11	11	14
Plain	**H**	**E**	**L**	**L**	**O**

Plaintext **HELLO** is recovered.

Why the key must be as long as the message

There is no repeating pattern. Letter 1 uses key letter 1, letter 2 uses key letter 2, and so on. If you reuse a short key, or reuse the same pad on a second message, the cipher stops being a one-time pad and can be attacked.

In the program, spaces are ignored. Only letters are added to the key.

Lab 6, Question 2 — Affine Cipher

What it is

Affine Cipher encrypts **one letter at a time** with a straight-line formula. The key is two numbers, a and b.

$$C = (a \cdot P + b) \bmod 26$$

$$P = a^{-1} \cdot (C - b) \bmod 26$$

a is the multiplier. b is the shift. a^{-1} is the modular inverse of a.

The rule for a

a must be **coprime with 26**: $\gcd(a, 26) = 1$. Otherwise a^{-1} does not exist and decryption is impossible.

Allowed values of a:

1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25

b can be any integer. It is used modulo 26, so 0 to 25 is enough.

Caesar Cipher is the special case $a = 1$. Then $C = (P + b) \bmod 26$.

Worked example

Use $a = 5$, $b = 8$, plaintext **HELLO**.

Letter	P	$5P + 8$	mod 26	Cipher
H	7	$35 + 8 = 43$	17	**R**
E	4	$20 + 8 = 28$	2	**C**

Letter	P	$5P + 8$	mod 26	Cipher
L	11	$55 + 8 = 63$	11	**L**
L	11	63	11	**L**
O	14	$70 + 8 = 78$	0	**A**

Ciphertext: **RCLLA**

Notice both **L**s became **L**. In Affine (and Caesar, and OTP with a repeated key letter) the same plaintext letter always becomes the same ciphertext letter. Hill is different: a letter's cipher depends on its partner in the pair.

Finding a^{-1}

We need $5x \equiv 1 \pmod{26}$.

$5 \times 21 = 105$, and $105 \bmod 26 = 1$ because $26 \times 4 = 104$. So $a^{-1} = 21$.

Decrypt RCLLA

$$P = 21 \cdot (C - 8) \bmod 26$$

First letter **R** = 17:

$$21 \cdot (17 - 8) = 21 \cdot 9 = 189, \quad 189 \bmod 26 = 7 \rightarrow \mathbf{H}$$

The same steps on C, L, L, A give E, L, L, O. Plaintext **HELLO** is recovered.

Side-by-side

Hill Cipher (Lab 5)

- Unit: 2 letters
- Key: a 2×2 matrix
- Encrypt: $C = K \times P \bmod 26$
- Decrypt: $P = K^{-1} \times C \bmod 26$
- Condition: determinant of K must be coprime with 26
- Lab result: MEETATBASE becomes **WSRZFRDCOE**

One Time Pad (Lab 6)

- Unit: 1 letter
- Key: one key letter for every plaintext letter
- Encrypt: $C = (P + K) \bmod 26$
- Decrypt: $P = (C - K) \bmod 26$
- Condition: key length equals message length, and the key is used once
- Lab result: HELLO with key XMCKL becomes **EQNVZ**

Affine Cipher (Lab 6)

- Unit: 1 letter

- Key: two numbers, a and b
- Encrypt: $C = (aP + b) \bmod 26$
- Decrypt: $P = a^{-1}(C - b) \bmod 26$
- Condition: a must be coprime with 26
- Lab result: HELLO with $a = 5$, $b = 8$ becomes **RCLLA**

What to say if asked in the lab

Hill: "We turn the message into pairs of numbers, multiply each pair by the key matrix modulo 26, and turn the results back into letters. To decrypt we multiply by the inverse matrix."

One Time Pad: "We add each plaintext letter to its own key letter modulo 26. Decryption subtracts the same key letter."

Affine: "We compute $(a \times \text{letter} + b) \bmod 26$. Decryption multiplies by the inverse of a after subtracting b. a must be coprime with 26."